

Adaptive Kernel Selection for Stein Variational Gradient Descent

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Abstract

Efficient posterior approximation remains a central problem in Bayesian inference. Stein Variational Gradient Descent (SVGD) is a deterministic variational inference method that transports particles toward a target distribution, but its practical success is governed by the reproducing kernel Hilbert space (RKHS) in which the transport direction is chosen. Kernel choice therefore determines both convergence behavior and approximation quality. The standard median heuristic provides a convenient bandwidth rule, yet it is too rigid for many problems and frequently breaks down in high-dimensional settings. We introduce a principled alternative: Adaptive SVGD (Ad-SVG), which selects kernel parameters from an abstract kernel family by maximizing the kernelized Stein discrepancy (KSD) during inference. The resulting algorithm alternates between SVGD particle transport and gradient-ascent updates of the kernel bandwidths, directly exploiting the discrepancy that controls the instantaneous SVGD descent direction. We give a streamlined theoretical analysis that extends fixed-kernel KSD convergence results to adaptive kernel selection and proves convergence of the maximal KSD over the kernel class. Across the experiments, Ad-SVG decisively improves over standard heuristic bandwidth selection while preserving the original SVGD framework.

1 Introduction

Stein Variational Gradient Descent (SVGD) [18] is a deterministic particle-based method for approximate Bayesian inference and a compelling alternative to traditional Markov Chain Monte Carlo (MCMC) methods. SVGD transports particles using update directions derived from the functional gradient of the Kullback-Leibler (KL) divergence to the target distribution, with the admissible directions constrained by a reproducing kernel Hilbert space (RKHS). This RKHS constraint makes kernel choice a first-order design decision rather than a secondary implementation detail: the selected kernel strongly shapes particle interactions and algorithmic performance [8, 28]. When this choice is poor, SVGD particle systems commonly underestimate posterior variance [2]. These failures have encouraged the view that SVGD is intrinsically fragile as the state dimension grows. We take a sharper position: a substantial part of this fragility comes from static and heuristic kernel selection. We therefore introduce an adaptive mechanism that tunes kernel parameters during inference by maximizing the kernelized Stein discrepancy (KSD), enabling the original SVGD transport mechanism to remain effective in more complex and higher-dimensional settings.

1.1 Related work.

Since its introduction [18], SVGD has become an established tool for approximate Bayesian inference across machine learning applications [22, 24, 30, 15]. The accompanying theory has also

matured substantially. Mean-field convergence is now understood in continuous-time [23, 8, 5] and discrete-time [16, 32] regimes, and finite-particle convergence rates have been proved under several assumption sets [3, 33]. A complementary line of work identifies SVGD dynamics as gradient flows on probability distribution spaces [19, 8], in close analogy with the variational formulation of Jordan, Kinderlehrer, and Otto [14].

Kernel selection is the central unresolved practical lever in SVGD because it determines particle interaction, descent geometry, and the strength of available convergence statements (see also Figures 1 and 9). Existing convergence results often establish mean-field convergence in KSD, and the implication from KSD control to weak convergence depends explicitly on the chosen kernel [11]. The median heuristic [12] remains popular because it is simple, but it offers no principled adaptation mechanism and is known to deteriorate as dimension increases [2, 40, 36]. Several recent methods address this high-dimensional degradation by modifying the SVGD geometry or update rule [7, 37, 10, 21]. Other approaches learn the update direction with neural networks [17, 39].

The closest precedent is Ai et al. [1], which develops a mixture-of-kernels framework. Their method defines a KSD for a weighted linear combination of kernels and learns the mixture weights by maximizing this multiple-kernel KSD. Our work moves beyond finite kernel bases: we optimize continuous kernel parameters directly, turning kernel selection into an adaptive component of the SVGD iteration rather than a discrete mixture-weight selection problem.

1.2 Contributions.

This paper establishes that SVGD can recover much of its practical strength without architectural additions when its kernel is selected principledly and adaptively. By isolating kernel selection from other SVGD modifications, we identify kernel choice as a decisive driver of performance and show how far adaptive kernel learning can push the original method itself.

Our main contributions are as follows.

- (i) **Adaptive Kernel Selection Method.** We introduce a KSD-driven kernel selection method that updates kernel parameters during SVGD inference. Unlike the median heuristic, which commits to a single scalar bandwidth rule, our approach optimizes multiple continuous kernel parameters and gives SVGD a substantially more expressive and responsive interaction geometry.
- (ii) **Theoretical Analysis.** We analyze the algorithm in the discrete-time mean-field setting and extend fixed-kernel convergence theory to adaptive kernel choice. In particular, we prove that the supremum of the KSD over the parameterized kernel class converges to zero as the particle distribution approaches the target. Under a Stein logarithmic Sobolev inequality, we further derive mean-field iteration-complexity bounds.
- (iii) **Empirical Validation.** Through numerical experiments, we show that adaptive kernel selection consistently improves on the median heuristic and directly alleviates variance collapse.

2 Mathematical background

We fix notation by taking a symmetric positive definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ with associated RKHS $\mathcal{H}_0$. The SVGD vector fields live in the d -fold product space $\mathcal{H}$, equipped with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$ and the canonical feature map $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. We write

$x \cdot y$ for the Euclidean inner product and $\nabla \cdot$ for divergence. The space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is $\mathcal{P}(\mathbb{R}^d)$, and $\mathcal{P}_p(\mathbb{R}^d)$ denotes those measures with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, the Wasserstein p -distance is

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of couplings of μ and ν , i.e. the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [26] quantify discrepancies between measures by maximizing the deviation of test-function integrals over a prescribed function class. Stein’s method [35] makes this strategy effective for intractable targets by producing test functions with zero target expectation. Specifically, for $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable kernel, $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [6, 20]. This construction yields the *kernelized Stein discrepancy* (KSD), defined by

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ is the unit ball in $\mathcal{H}$. The optimizer is explicit: $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, where $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$, and Φ_k is the feature map of k . Consequently, the supremum is exactly the RKHS norm of ψ , giving the equivalent characterization $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$.

2.2 Stein Variational Gradient Descent.

Given a target distribution π and reference distribution μ_0 , SVGD constructs an approximation of π by repeatedly pushing μ_0 forward: $\mu_{n+1} := T_{\#} \mu_n$, where $T_{\#} \cdot$ denotes push-forward under $T(x) = x + \gamma \psi^{\mu_n}(x)$. The vector field $\psi^{\mu} = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$ for $\mu \in \mathcal{P}(\mathbb{R}^d)$ is the RKHS steepest-descent direction. The KSD variational problem in Equation (1) makes this descent interpretation precise: $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}} \min_{\gamma} \frac{d}{d\gamma} \text{KL}(T_{\#} \mu_n \| \pi) \Big|_{\gamma=0}$ over the unit ball of $\mathcal{H}$ [cf. 18, Theorem 3.1], where $\text{KL}(\cdot \| \cdot)$ denotes the KL-divergence. In particular, this choice gives

$$\frac{d}{d\gamma} \text{KL} \left((\text{Id} + \gamma \psi^{\mu_n})_{\#} \mu_n \| \pi \right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n|\pi). \quad (2)$$

Applying these maps to particles $\{X_0^i\}_{i=1}^M$ sampled from μ_0 yields the standard SVGD particle update:

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because KL convergence for SVGD is difficult to prove and usually requires restrictive assumptions, recent theory has focused on convergence in kernelized Stein discrepancy (KSD) [16, 33, 32]. This

shift makes kernel choice even more consequential. KSD minimization alone does not automatically imply weak convergence: the mean-field measures may fail to be tight, and tightness is necessary for weak convergence in Polish spaces. Moreover, Equation (1) shows that the RKHS structure, hence the kernel, controls how KSD convergence relates to weak convergence [11].

We therefore formulate SVGD over a parameterized kernel family $\{k_\theta \mid \theta \in \Theta\}$ and make kernel selection part of the algorithm. For each $\theta \in \Theta$, KSD_θ denotes the corresponding *kernelized Stein discrepancy*, ψ_θ^μ the optimal update direction, and Φ_θ the associated feature map. This parameterization exposes the kernel as an optimizable object and directly addresses the limitations of committing to a fixed kernel before inference begins.

The key principle is direct: select the kernel that maximizes the KSD between the current empirical particle distribution and the target. Under SVGD, the instantaneous decrease in KL divergence is proportional to the squared KSD at the current particle measure (see Equations (2) and (7)). Hence, at each iteration, maximizing KSD selects the kernel that maximizes the available instantaneous KL descent within the candidate class. Geometric analyses of SVGD as a gradient flow support the same conclusion: kernels producing larger KSD values induce more favorable convergence behavior when the associated RKHSs are compared [28, 8].

Where vanilla SVGD fixes a kernel and studies convergence of the corresponding KSD, Ad-SVGd controls the maximal KSD over the kernel class. Unlike the median heuristic used by Liu and Wang [18], the method applies to any parameterized kernel family: at each step, it targets an optimal parameter

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

In practice, we update the kernel parameter by one or more gradient-ascent steps on the KSD with step size $s > 0$ before the particle update, and we allow the update to be skipped on selected SVGD iterations to reduce runtime (see Algorithm 1). The user-specified variable `paramupdate` implements this choice. The particle-transport step itself can be replaced by stronger SVGD variants, including adaptive step-size schedules, line searches, momentum, or second-order/preconditioned updates, without changing the kernel-selection mechanism. Likewise, the gradient-ascent step for the kernel can be replaced by another optimization routine. Our implementation uses the identity [6, 20]

$$\text{KSD}^2(\mu | \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k d(\mu \otimes \mu) \quad (4)$$

with

$$\begin{aligned} u_\pi^k(x, y) = & k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ & + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \end{aligned} \quad (5)$$

which gives the required kernel-parameter gradients directly. The additional cost of Ad-SVGd is controlled by two concrete features:

- Kernel-parameter gradient ascent reuses the same gradients $\nabla \log \pi(X_n^i)$ computed for the SVGD step, so Ad-SVGd does not require extra evaluations of $\log \pi$ gradients.
- The overhead can be reduced by using only a small number of kernel updates `nstepstheta`, by updating the kernel only after several SVGD steps, or, for large particle systems, by estimating the KSD with subsampled particles.

Algorithm 1: Ad-SVGD

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps $nsteps, nstepstheta \in \mathbb{N}$

Output: Final particle set $\{X_{nsteps}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```
for  $n = 0$  to  $nsteps - 1$  do
    Decide paramupdate  $\in \{\text{True}, \text{False}\}$ ;
    if paramupdate then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to  $nstepstheta - 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{nstepstheta}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end
```

4 Convergence analysis

We now analyze a target measure with Lebesgue density $\pi(x) \propto \exp(-V)$ for a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. The analysis shows that adaptive kernel selection fits cleanly into recent KSD convergence theory for SVGD. Specifically, we extend the discrete-time analysis of [32] under Talagrand’s inequality, a condition that holds under mild assumptions on the target and is weaker than the logarithmic Sobolev inequality commonly used in related analyses. We impose the assumptions of [32] uniformly over all kernel parameters.

Assumption 1. We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.

Assumption 2. We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand’s inequality T1, which means that there exists $\lambda > 0$ such that

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \parallel \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.

4.1 Convergence analysis of SVGD under Talagrand’s inequality

In the following, we denote

$$\mathcal{F}(\mu) := \text{KL}(\mu \parallel \pi).$$

and use the following fundamental inequality. Fix a kernel parameter $\theta \in \Theta$ satisfying Assumption 3, and define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for arbitrary $g \in \mathcal{H}$. Under Assumptions 1 and 3, take $\gamma, B > 0$, $\alpha > 1$, and $g \in \mathcal{H}_\theta$ such that $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$. Proposition 3.1 of [32] gives

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta} \quad (6)$$

with $K = (\alpha^2 + L)B$. Applying this inequality to the fixed-kernel iterates (3) yields the descent condition of [32, Theorem 3.2],

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1}. \quad (8)$$

The decisive step is verifying $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$ via Talagrand's inequality, Assumption 2, which permits (6) with $g = \psi_\theta^{\mu_n}$. The descent condition (7) then proves

$$\lim_{n \rightarrow \infty} \text{KSD}_\theta^2(\mu_n | \pi) = 0,$$

because $\sum_{n=0}^{\infty} \text{KSD}_\theta^2(\mu_n | \pi) \leq c_\gamma^{-1} \mathcal{F}(\mu_0)$ for $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$.

With adaptive kernel parameters (θ_n) , the same mechanism gives the stepwise descent bound

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_{\theta_n}^2(\mu_n | \pi).$$

4.2 Convergence analysis of Ad-SVGd.

First consider the idealized adaptive SVGd iteration

$$\begin{aligned} \mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n}) \# \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi), \end{aligned} \quad (9)$$

so that

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

This formulation is exact whenever the KSD maximization over kernel parameters admits a closed-form solution, for example in the multiple-kernel SVGd framework [1], where the kernel is a convex combination of base kernels

$$k_\theta(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

For general continuous kernel families, the update θ_n in (9) must be computed approximately. Throughout this setting, we assume that $\operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

The resulting descent condition follows directly from Theorem 3.2 of [32]:

Lemma 1. Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

there exists $c_\gamma > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi) = 0$.

The practically relevant case is approximate KSD maximization over the kernel parameter. Assume $\max_{\theta \in \Theta} \text{KSD}_\theta(\mu | \pi) < \infty$ for any $\mu \in \mathcal{P}(\mathbb{R}^d)$ and consider the alternating scheme

$$\begin{aligned} \theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\text{Id} + \gamma \psi_{\theta_n}^{\mu_n})_\# \mu_n \end{aligned} \tag{10}$$

where $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ is an iterative update rule designed to maximize $\text{KSD}_\theta(\mu_n | \pi)$. In Algorithm 1, Ψ_n is the gradient-ascent update for the KSD, but the analysis also covers other optimization schemes that deliver the same approximation property. The required behavior is stated next.

Assumption 4. We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and

$$\max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi) - \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

This approximation condition is sufficient for the main adaptive convergence guarantee.

Theorem 3. Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [32] we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_\gamma \text{KSD}_{\theta_n}^2(\mu_n | \pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and, in particular, it is a Cauchy sequence. Hence, we have

$$c_\gamma \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n | \pi) \leq \text{KSD}_{\theta_n}^2(\mu_n | \pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

Under a Stein logarithmic Sobolev inequality, the result strengthens from maximal-KSD convergence to KL convergence. For several regimes of ε_n , we obtain iteration-complexity bounds for the mean-field system (10); the same analysis quantifies the bias induced by solving the KSD maximization only to a fixed $\bar{\varepsilon}$ accuracy at each iteration. Section B gives the details. Importantly, the condition only requires one kernel in the class $\{k_\theta\}$ to satisfy the Stein logarithmic Sobolev inequality, which makes the assumption substantially more flexible than a uniform requirement over the entire class.

5 Numerical Experiments

We evaluate Ad-SVGd on two main numerical experiments: a one-dimensional Gaussian mixture and a linear inverse problem governed by an ordinary differential equation. Section A adds controlled toy experiments that isolate the effect of increasing dimension. Section 5.4 further demonstrates the advantage of Ad-SVGd over median-heuristic SVGd on Bayesian logistic regression, a benchmark used in prior SVGd work.

5.1 Kernel parameterization

SVGd is typically implemented with kernels of the form

$$k_h(x, y) = \exp\left(-\frac{\|x - y\|_p^p}{h}\right),$$

where $\|\cdot\|_p$ is the p -norm on $\mathbb{R}^d$ [e.g. 18, 2, 8]. The crucial parameter is the kernel *bandwidth* h . The median heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, where med is the current median particle distance, and is motivated by balancing the two terms in the SVGd update (3) [18].

To exploit the flexibility of adaptive selection, we use product kernels

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

with parameter $h = (h_1, \dots, h_d)$, allowing each dimension to have its own bandwidth. The derivatives required by Algorithm 1 for these kernels follow directly from Equations (4) and (5). We also tested a dimension-wise median heuristic for these product kernels; it consistently failed to deliver the desired behavior and exhibited variance collapse.

The same Ad-SVGd mechanism immediately supports richer and more scalable kernel families. Spectral or low-rank parameterizations can adapt dominant variability directions without the exponential cost of full product kernels, while Matérn and mixed-product constructions can encode smoothness and anisotropy more explicitly.

5.2 Toy example

We first revisit the one-dimensional example of Liu and Wang [18], a two-component Gaussian mixture $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. We use $M \in \{50, 200, 500\}$ particles initialized from

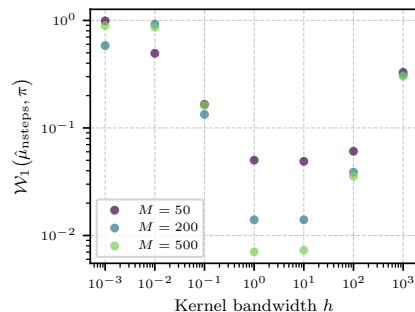


Figure 1: Final Wasserstein 1 distances in the

$\mathcal{N}(0, 1)$ and run SVGD for 10^4 steps with step size 1, using the kernels above with $p = 1$ and fixed bandwidths h . We measure sample quality with the Wasserstein 1-distance $\mathcal{W}_1$, computed from the explicit formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_\mu(x) - F_\nu(x)| dx$ (see Panaretos and Zemel [29]) and an exact sample of size 10^5 . Figure 1 makes the central point clear: SVGD performs well only over a limited bandwidth range. Even in this simple setting, the method is highly sensitive to h , so kernel selection is essential rather than optional.

5.3 Linear inverse problem based on ODE

The next experiment, adapted from [38, Example 2.1], tests whether adaptive kernel selection improves uncertainty quantification in a structured inverse problem. We consider the one-dimensional differential equation

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

and the associated inverse problem of recovering the right-hand side $u(\cdot) \in L^2([0, 1])$ from discrete noisy observations of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. The observation model is

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ is observational noise and the forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is $\mathcal{O} \circ H^{-1}$. Here $H(f) = -f'' + f$, and $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ observes the solution at N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian inverse problem, we place a Gaussian process (GP) prior on u through the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

where $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and the coefficients are independent with $x_k \sim \mathcal{N}(0, \lambda_k)$ and $\lambda_k = 50k^{-2}$. In this formulation, inference reduces to estimating the KL coefficients $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$ under the Gaussian prior $\mathcal{N}(0, \Gamma_0)$, where $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. With $\varepsilon \sim \mathcal{N}(0, \Gamma)$, the posterior density is

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

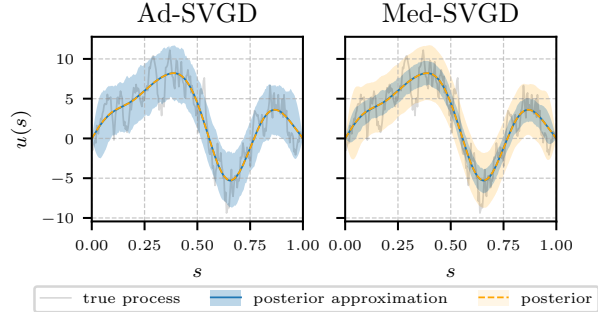


Figure 2: GP reconstruction for the ODE-based inverse problem, comparing posterior means and 90% confidence intervals.

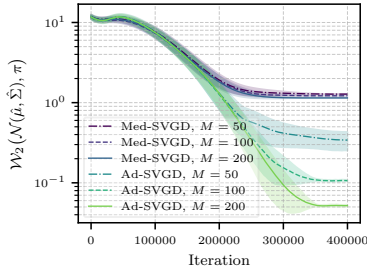
For implementation, we discretize H in Equation (11) on a grid with mesh size 2^{-8} and use the fully observed system, so $N_{\text{obs}} = 2^8$. We take $N_x = 16$ KL terms for u and set the noise covariance to $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. Reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

We compare median-heuristic SVGD (Med-SVGd) with Ad-SVGd for particle ensemble sizes $M \in \{50, 100, 200\}$. Both methods use the kernels from Section 5.1 with $p = 1$. We run $4 \cdot 10^5$ iterations with particle step size 10^{-3} and, following Liu and Wang [18], an AdaGrad variant for adaptive step-size control. For Ad-SVGd, the bandwidth step size is 10^{-5} , and the bandwidth is updated once every 100 particle updates. Under this schedule, adaptive kernel selection adds no significant runtime difference relative to Med-SVGd.

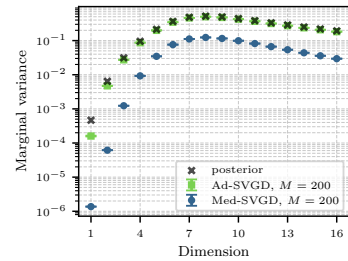
Figure 2 shows a representative GP reconstruction. Both methods approximate the posterior mean well, but Ad-SVGd is the method that captures posterior uncertainty accurately. To quantify approximation quality, we compute the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the Gaussian $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ induced by the particle sample mean $\hat{\mu}$ and sample covariance $\hat{\Sigma}$ [29, Section 3.2]. Since π is multivariate normal, this distance has the explicit form

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π are the posterior mean and covariance. We also compare final marginal particle variances against the posterior variances. Figure 3 aggregates the results over 56 random seeds; the posterior covariance is independent of $\bar{y}$. The results consistently favor Ad-SVGd: Med-SVGd underestimates uncertainty, whereas Ad-SVGd gives a substantially better posterior approximation. Moreover, Ad-SVGd continues to improve when the particle count increases beyond 50, while Med-SVGd does not show the same benefit. Figure 4 displays the learned bandwidths. The component-wise bandwidths stabilize earlier than the approximation error, differ clearly across components, and end up negatively correlated with the corresponding marginal variances.



(a) Convergence of approximation error

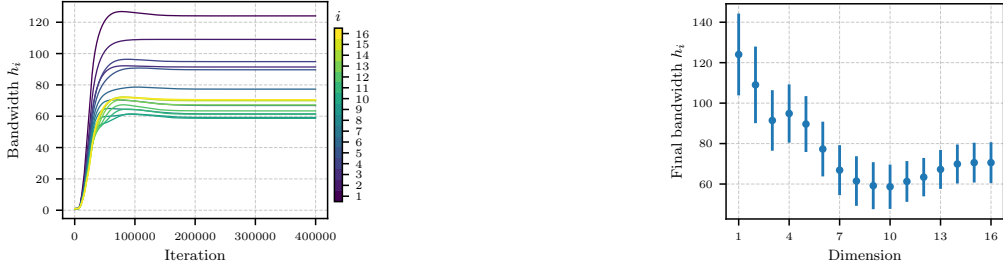


(b) Marginal variances

Figure 3: Aggregated ODE inverse-problem results for Med-SVGd and Ad-SVGd, reported as mean and 95% confidence interval over 56 random seeds.

5.4 Bayesian logistic regression

Following Liu and Wang [18] and Liu et al. [21], we evaluate Bayesian logistic regression (BLR) on the Covertypes data set [4]. The regression weights w have prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$, and the target is the posterior of $x = [w, \alpha]$. As in Liu et al. [21], we use the



(a) Evolution of bandwidth per dimension (mean) (b) Final bandwidths (mean and 95% conf. interval)

Figure 4: Ad-SVGD bandwidth behavior on the ODE inverse problem, aggregated over 56 random seeds.

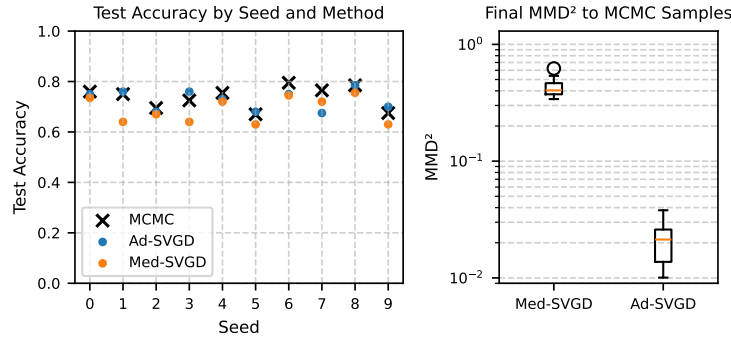


Figure 5: Bayesian logistic-regression approximation quality, measured by prediction accuracy (left) and MMD^2 to NUTS reference samples (right).

No-U-Turn Sampler (NUTS) of Hoffman and Gelman [13] to construct a reference against which Med-SVGD and Ad-SVGD are evaluated. Following Liu et al. [21], we run the methods on random subsets of 1000 data points and aggregate over 10 draws. Classification performance is measured by test-set prediction accuracy using the particle mean. Figure 5 shows that Ad-SVGD outperforms Med-SVGD for almost every seed, achieving higher predictive accuracy and a posterior approximation closer to the MCMC reference.

The same figure reports posterior approximation accuracy through the squared maximum mean discrepancy (MMD^2) [25, Section 3.5] to the MCMC samples. The gap between the methods is substantial. Figure 7 confirms this conclusion by comparing final-particle covariance matrices with the MCMC reference, averaged over the 10 seeds and visualized as in Liu et al. [21]. Med-SVGD severely underestimates posterior uncertainty, whereas Ad-SVGD tracks the covariance structure much more

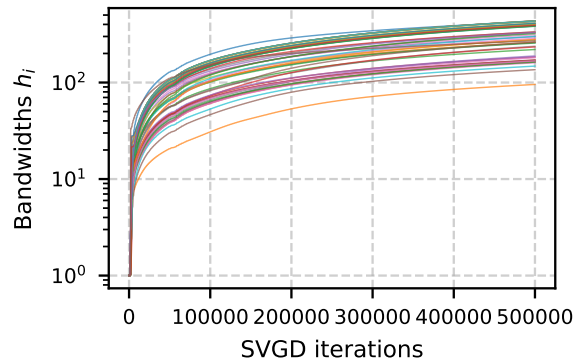


Figure 6: Learned bandwidth evolution in the BLR experiment.

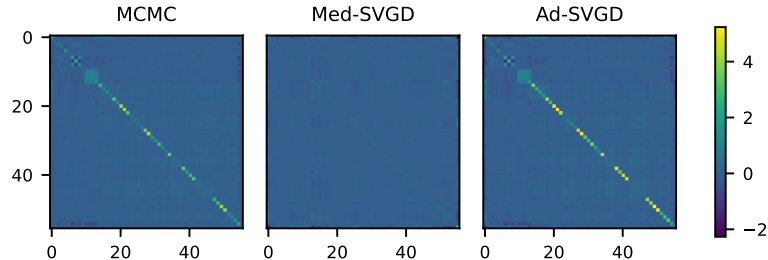


Figure 7: Covariance matrices of final SVGD particle sets against MCMC reference samples, averaged over 10 seeds.

faithfully. Figure 6 shows why the adaptive mechanism matters: starting from an uninformed initialization of 1 in every component, Ad-SVGD recovers bandwidths that differ from the initialization by several orders of magnitude.

6 Limitations and outlook

The main theoretical limitation is the reliance on Assumption 4. Our ongoing work studies when the alternating gradient-ascent scheme in Algorithm 1 satisfies this assumption. Even without a general guarantee for the implemented optimizer, the experiments show that the assumption captures a practically effective mechanism: the resulting Ad-SVGD implementation delivers strong empirical gains.

Our analysis deliberately focuses on the original SVGD dynamics, which isolates the contribution of adaptive kernel selection. This makes the method directly complementary to recent variants such as sliced SVGD [10], Grassmann SVGD [21], and Stein transport [27]; combining these approaches with Ad-SVGD is a natural next step.

References

- [1] Qingzhong Ai et al. “Stein Variational Gradient Descent with Multiple Kernels”. In: *Cognitive Computation* 15.2 (2023), pp. 672–682. DOI: [10.1007/s12559-022-10069-5](https://doi.org/10.1007/s12559-022-10069-5).
- [2] Jimmy Ba et al. “Understanding the Variance Collapse of SVGD in High Dimensions”. In: *International Conference on Learning Representations*. 2022.
- [3] Krishna Balasubramanian, Sayan Banerjee, and Promit Ghosal. “Improved Finite-Particle Convergence Rates for Stein Variational Gradient Descent”. In: *The Thirteenth International Conference on Learning Representations*. 2025.
- [4] Jock Blackard. *Coverttype*. UCI Machine Learning Repository. 1998. DOI: [10.24432/C50K5N](https://doi.org/10.24432/C50K5N).
- [5] Sinho Chewi et al. “SVGD as a kernelized Wasserstein gradient flow of the chi-squared divergence”. In: *Advances in Neural Information Processing Systems* 33. 2020.
- [6] Kacper Chwialkowski, Heiko Strathmann, and Arthur Gretton. “A Kernel Test of Goodness of Fit”. In: *Proceedings of The 33rd International Conference on Machine Learning*. 2016.
- [7] Gianluca Detommaso et al. “A Stein variational Newton method”. In: *Advances in Neural Information Processing Systems* 31. 2018.
- [8] Andrew Duncan, Nikolas Nüsken, and Lukasz Szpruch. “On the geometry of Stein variational gradient descent”. In: *Journal of Machine Learning Research* 24 (2023).
- [9] Rémi Flamary et al. “POT: Python Optimal Transport”. In: *Journal of Machine Learning Research* 22 (2021).
- [10] Wenbo Gong, Yingzhen Li, and José Miguel Hernández-Lobato. “Sliced Kernelized Stein Discrepancy”. In: *International Conference on Learning Representations*. 2021.
- [11] Jackson Gorham and Lester Mackey. “Measuring Sample Quality with Kernels”. In: *Proceedings of the 34th International Conference on Machine Learning*. 2017.
- [12] Arthur Gretton et al. “A Kernel Two-Sample Test”. In: *Journal of Machine Learning Research* 13 (2012).
- [13] Matthew D. Hoffman and Andrew Gelman. “The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo”. In: *Journal of Machine Learning Research* 15 (2014).
- [14] Richard Jordan, David Kinderlehrer, and Felix Otto. “The Variational Formulation of the Fokker–Planck Equation”. In: *SIAM Journal on Mathematical Analysis* 29.1 (1998), pp. 1–17. DOI: [10.1137/S0036141096303359](https://doi.org/10.1137/S0036141096303359).
- [15] Rahif Kassab and Osvaldo Simeone. “Federated Generalized Bayesian Learning via Distributed Stein Variational Gradient Descent”. In: *IEEE Transactions on Signal Processing* 70 (2022), pp. 2180–2192. DOI: [10.1109/TSP.2022.3168490](https://doi.org/10.1109/TSP.2022.3168490).
- [16] Anna Korba et al. “A Non-Asymptotic Analysis for Stein Variational Gradient Descent”. In: *Advances in Neural Information Processing Systems* 33. Vol. 33. 2020.
- [17] Lauro Langosco di Langosco, Vincent Fortuin, and Heiko Strathmann. “Neural Variational Gradient Descent”. In: *Fourth Symposium on Advances in Approximate Bayesian Inference*. 2022.

- [18] Q. Liu and D. Wang. “Stein Variational Gradient Descent: A General Purpose Bayesian Inference Algorithm”. In: *Advances in Neural Information Processing Systems 29*. 2016.
- [19] Qiang Liu. “Stein Variational Gradient Descent as Gradient Flow”. In: *Advances in Neural Information Processing Systems 30*. 2017.
- [20] Qiang Liu, Jason Lee, and Michael Jordan. “A Kernelized Stein Discrepancy for Goodness-of-fit Tests”. In: *Proceedings of The 33rd International Conference on Machine Learning*. 2016.
- [21] Xing Liu et al. “Grassmann Stein Variational Gradient Descent”. In: *Proceedings of The 25th International Conference on Artificial Intelligence and Statistics*. 2022.
- [22] Yang Liu et al. “Stein Variational Policy Gradient”. In: *Proceedings of the Thirty-Third Conference on Uncertainty in Artificial Intelligence*. 2017.
- [23] Jianfeng Lu, Yulong Lu, and James Nolen. “Scaling Limit of the Stein Variational Gradient Descent: The Mean Field Regime”. In: *SIAM Journal on Mathematical Analysis* 51.2 (2019), pp. 648–671. DOI: [10.1137/18M1187611](https://doi.org/10.1137/18M1187611).
- [24] Safa Messaoud et al. “S2AC: Energy-Based Reinforcement Learning with Stein Soft Actor Critic”. In: *The Twelfth International Conference on Learning Representations*. 2024.
- [25] Krikamol Muandet et al. “Kernel Mean Embedding of Distributions: A Review and Beyond”. In: *Foundations and Trends in Machine Learning* 10.1-2 (2017), pp. 1–141. DOI: [10.1561/22000000060](https://doi.org/10.1561/22000000060).
- [26] Alfred Müller. “Integral Probability Metrics and Their Generating Classes of Functions”. In: *Advances in Applied Probability* 29.2 (1997), pp. 429–443. DOI: [10.2307/1428011](https://doi.org/10.2307/1428011).
- [27] Nikolas Nüsken. *Stein transport for Bayesian inference*. 2024. arXiv: [2409.01464](https://arxiv.org/abs/2409.01464).
- [28] Nikolas Nüsken and D. R. Michiel Renger. “Stein variational gradient descent: Many-particle and long-time asymptotics”. In: *Foundations of Data Science* 5.3 (2023), pp. 286–320. DOI: [10.3934/fods.2022023](https://doi.org/10.3934/fods.2022023).
- [29] Victor M. Panaretos and Yoav Zemel. “Statistical Aspects of Wasserstein Distances”. In: *Annual Review of Statistics and Its Application* 6 (2019), pp. 405–431. DOI: [10.1146/annurev-statistics-030718-104938](https://doi.org/10.1146/annurev-statistics-030718-104938).
- [30] Yuchen Pu et al. “VAE Learning via Stein Variational Gradient Descent”. In: *Advances in Neural Information Processing Systems 30*. 2017.
- [31] Sebastian Reich and Simon Weissmann. “Fokker–Planck Particle Systems for Bayesian Inference: Computational Approaches”. In: *SIAM/ASA Journal on Uncertainty Quantification* 9.2 (2021), pp. 446–482. DOI: [10.1137/19M1303162](https://doi.org/10.1137/19M1303162).
- [32] Adil Salim, Lukang Sun, and Peter Richtarik. “A Convergence Theory for SVGD in the Population Limit under Talagrand’s Inequality T1”. In: *Proceedings of the 39th International Conference on Machine Learning*. 2022.
- [33] Jiaxin Shi and Lester Mackey. “A Finite-Particle Convergence Rate for Stein Variational Gradient Descent”. In: *Advances in Neural Information Processing Systems 36*. 2023.
- [34] Björn Sprungk, Simon Weissmann, and Jakob Zech. “Metropolis-adjusted interacting particle sampling”. In: *Statistics and Computing* 35.64 (2025). DOI: [10.1007/s11222-025-10595-w](https://doi.org/10.1007/s11222-025-10595-w).

- [35] Charles Stein. “A bound for the error in the normal approximation to the distribution of a sum of dependent random variables”. In: *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, Volume 2: Probability Theory*. 1972, pp. 583–603.
- [36] Dilin Wang, Zhe Zeng, and Qiang Liu. “Stein Variational Message Passing for Continuous Graphical Models”. In: *Proceedings of the 35th International Conference on Machine Learning*. 2018.
- [37] Dilin Wang et al. “Stein Variational Gradient Descent With Matrix-Valued Kernels”. In: *Advances in Neural Information Processing Systems 32*. 2019.
- [38] Simon Weissmann, Ashia Wilson, and Jakob Zech. “Multilevel Optimization for Inverse Problems”. In: *Proceedings of Thirty Fifth Conference on Learning Theory*. 2022.
- [39] Qian Zhao et al. “Stein variational gradient descent with learned direction”. In: *Information Sciences* 637 (2023). DOI: [10.1016/j.ins.2023.118975](https://doi.org/10.1016/j.ins.2023.118975).
- [40] Jingwei Zhuo et al. “Message Passing Stein Variational Gradient Descent”. In: *Proceedings of the 35th International Conference on Machine Learning*. 2018.

A More numerical experiments

A.1 Gaussian mixture models

We revisit the setup from Section 5.2. For that experiment, Figure 9 compares final particle histograms with the target density for $h \in \{0.001, 1, 1000\}$, making the effect of bandwidth misspecification visible. We then compare Med-SVGD and Ad-SVGD directly by varying the number of particles M and evaluating the final particle distributions after 10^4 iterations with step size 1.

Figure 8 reports the Wasserstein 1-distance between the final empirical particle distribution and the target for Med-SVGD and Ad-SVGD with $M \in \{10, 20, 50, 100, 200, 500\}$. As expected, approximation quality improves with N for both methods. In this one-dimensional setting, both methods reach comparable accuracy, with Wasserstein distance below 0.01 for $N = 500$.

Figure 8: Approximation error for the 1D Gaussian mixture, comparing Med-SVGD and Ad-SVGD across particle counts.

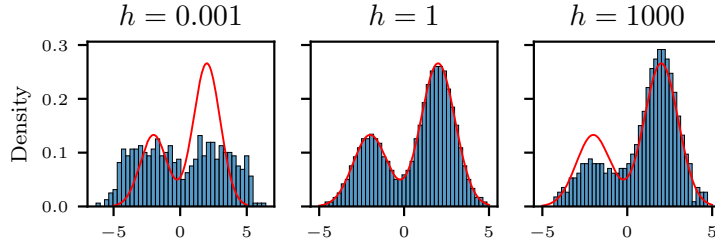
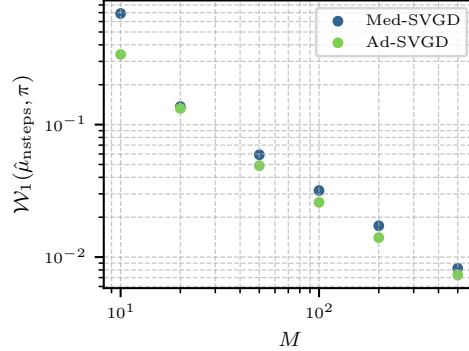


Figure 9: Final-particle histograms from fixed-bandwidth SVGD with $h = 0.001, 1, 1000$; the target density π is shown in red.

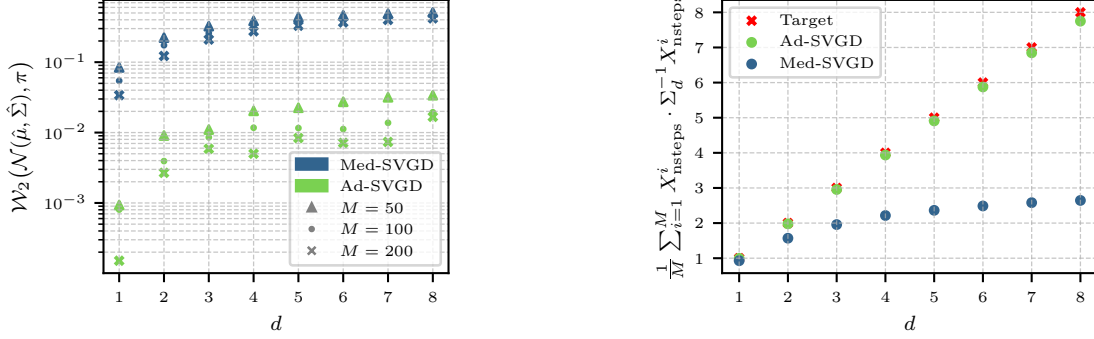
In the one-dimensional case, both methods approximate the target well. This confirms that Ad-SVGD preserves the strong behavior of the median heuristic in simple settings, while the adaptive mechanism becomes decisive in the higher-dimensional experiments that follow.

A.2 Scaling dimension: multivariate normal distribution

Moving to higher dimensions, we consider Gaussian targets $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$. This construction creates increasingly anisotropic targets while keeping the ground truth explicit. We initialize particles from $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$ and run the algorithms for 10^4 iterations with step size 0.1. When needed for numerical stability, we use a smaller step size and adjust the number of iterations accordingly.

We compare the final particle quality of Med-SVGD and Ad-SVGD as dimension increases. Since the Wasserstein 2-distance between Gaussian distributions has an explicit formula (see [29]),

we measure sample quality by the Wasserstein 2-distance between π_d and $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, where $\hat{\mu}$ and $\hat{\Sigma}$ are the empirical particle mean and covariance [29, Section 3.2]. Following the corresponding Python Optimal Transport terminology [9], we call this the *Bures Wasserstein distance*. Figure 10a shows this metric for Med-SVGD and Ad-SVGD with $M = 50, 100, 200$ as dimension increases. Ad-SVGD significantly outperforms Med-SVGD for every tested value of d .



(a) Approximation error for $M \in \{50, 100, 200\}$.

(b) χ^2 diagnostic statistic for $M = 200$.

Figure 10: Results for anisotropic multivariate normal targets of increasing dimension.

The Bures Wasserstein distance is relatively insensitive to the smallest target variances, so we also evaluate the statistic $X \cdot \Sigma_d^{-1} X$, which is χ^2 -distributed with d degrees of freedom and expectation d for $X \sim \mathcal{N}(0, \Sigma_d)$ (cf. [34, p. 27]). We compute the particle average of this statistic for $M = 200$ and compare it with the expected value d in Figure 10b. Beyond one dimension, the diagnostic exposes a clear failure mode: the Med-SVGD statistic drifts increasingly far from the true expectation as d grows, while the Ad-SVGD statistic stays close. Thus, adaptive kernel selection correctly handles these higher-dimensional anisotropic targets where the median heuristic does not.

Tables 1 and 2 gives a component-level view for $M = 200$ across dimensions. Each table lists marginal particle variances against the target marginal variances in the first row. Table 1 shows that Med-SVGD substantially underestimates uncertainty, whereas Table 2 shows that Ad-SVGD closely matches the target variances across components. This directly corroborates the diagnostic in Figure 10b.

Finally, we examine the marginal particle distributions for $M = 200$ and $d = 8$. For visualization, we normalize each component by the inverse target marginal standard deviation, equivalently multiplying the i -th component by i , so every target marginal becomes standard normal. Figures 11 and 12 show the normalized marginal histograms for Med-SVGD and Ad-SVGD, with a standard Gaussian density for reference. Ad-SVGD captures all marginals well, while Med-SVGD visibly underestimates uncertainty. The Q-Q plots in Figures 13 and 14 confirm the same conclusion by comparing normalized particle quantiles with standard-normal target quantiles.

Table 1: Marginal variances of final Med-SVGD particle distributions for the d -dimensional examples with $M = 200$, shown against the target variances of π_d .

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Marginal variances of final Ad-SVGD particle distributions for the d -dimensional examples with $M = 200$, shown against the target variances of π_d .

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

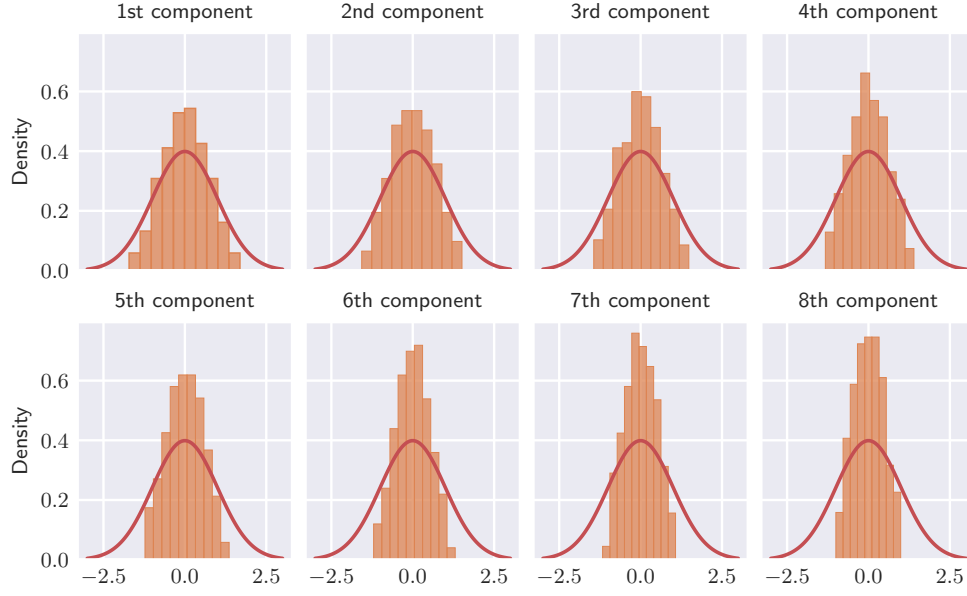


Figure 11: Component histograms for the final Med-SVGd particles in the eight-dimensional example with $M = 200$, rescaled by target marginal variances and compared with the standard Gaussian density.

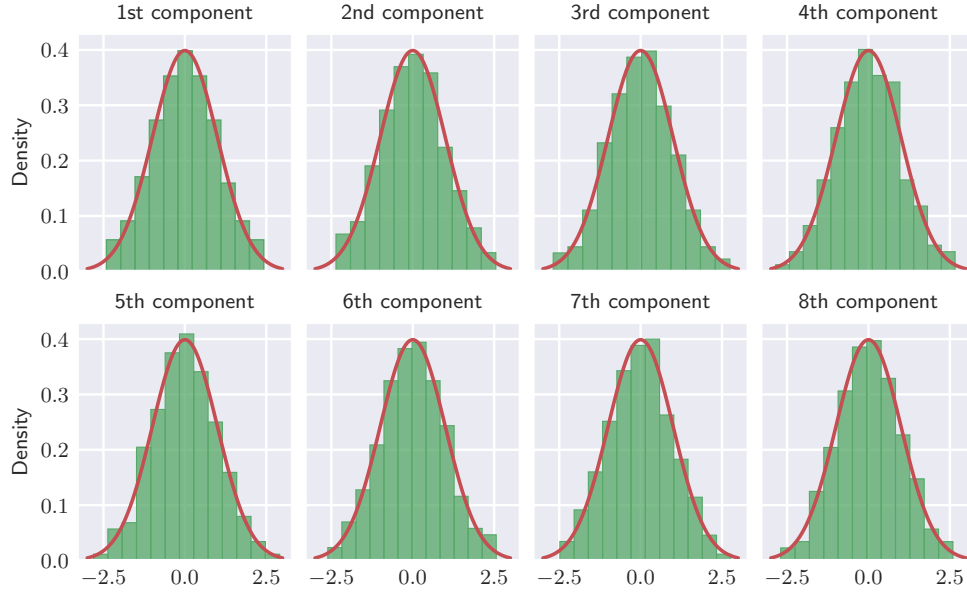


Figure 12: Component histograms for the final Ad-SVGd particles in the eight-dimensional example with $M = 200$, rescaled by target marginal variances and compared with the standard Gaussian density.

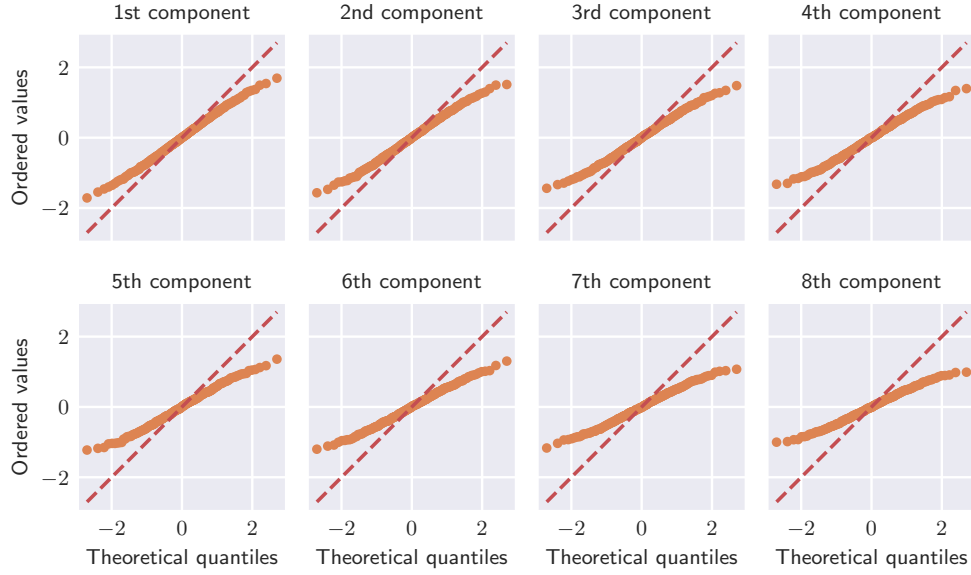


Figure 13: Q-Q plots for rescaled Med-SVGD marginals in the eight-dimensional example with $M = 200$, compared with a standard normal distribution and the unit-slope reference line through the origin.

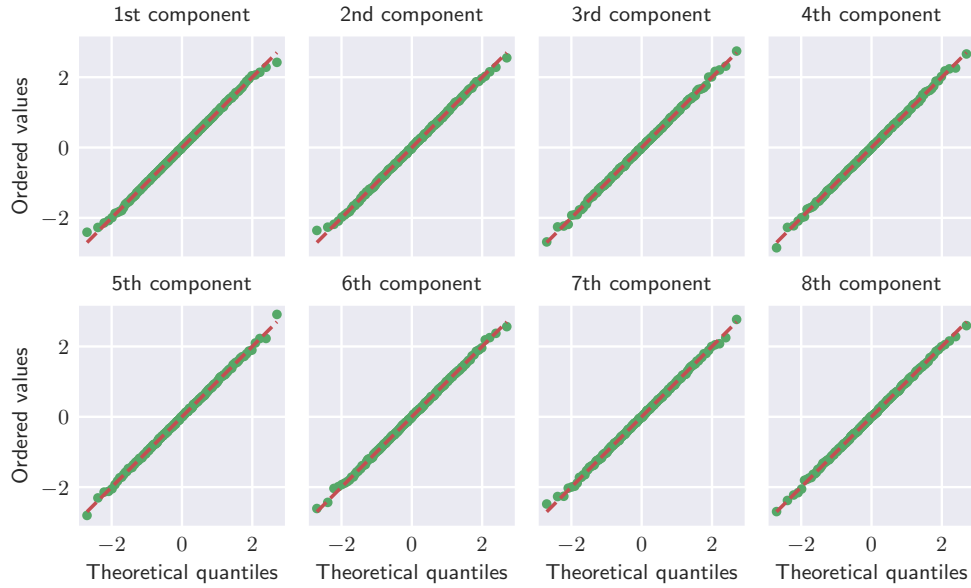


Figure 14: Q-Q plots for rescaled Ad-SVGD marginals in the eight-dimensional example with $M = 200$, compared with a standard normal distribution and the unit-slope reference line through the origin.

A.3 Gaussian process inference

We further test Ad-SVGD under dimension scaling in the Gaussian process (GP) inference task proposed by [31]. The GP on $[0, 1]$ is represented by the truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$ and independent Gaussian coefficients $x_k \sim \mathcal{N}(0, k^{-2})$. We observe the process at N_y equispaced points in $[0, 1]$ and infer the coefficients x_k . For fixed N_x and N_y , this is a linear inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$, $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with diagonal entries k^{-2} for $k = 1, \dots, N_x$, and the noise is independent Gaussian, $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. Reference observations $\bar{y} \in \mathbb{R}^{N_y}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

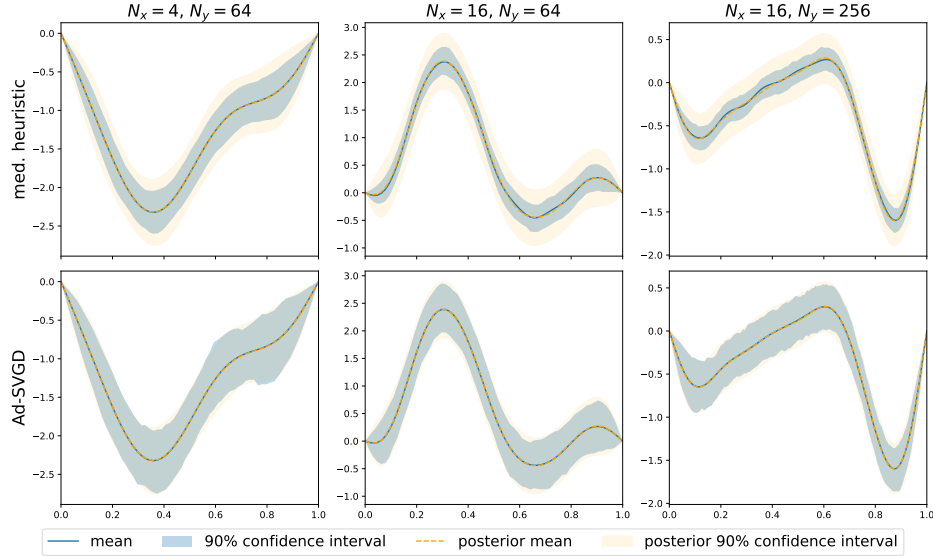


Figure 15: Estimated processes from the median heuristic and Ad-SVGD with $M = 100$ particles, compared with the posterior across model configurations.

We use SVGD to sample from the posterior $X | Y = \bar{y}$ and compare the median heuristic with adaptive kernel selection. Median-heuristic SVGD uses $k(x, y) = \exp(-\|x - y\|_1/h)$, while Ad-SVGD uses product kernels $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with $h = (h_1, \dots, h_{N_x})$, assigning a separate bandwidth to each dimension. Following [18], we use an Adagrad variant for step-size control when $N_x = 16$.

Figure 15 compares the resulting processes with the posterior for $N_x = 4$ and $N_y = 64$, $N_x = 16$ and $N_y = 64$, and $N_x = 16$ and $N_y = 256$. As dimension increases, median-heuristic SVGD underestimates posterior variance, whereas Ad-SVGD remains much closer to the posterior uncertainty. This pattern is stable across different numbers of GP observations. Table 3 quantifies the effect through the trace of the particle covariance relative to the true posterior covariance: the median heuristic severely underestimates uncertainty, while Ad-SVGD captures the variance substantially more accurately.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the final particle covariance ($M = 100$) compared with the theoretical posterior covariance across configurations, averaged over 25 runs.

B Refined theoretical analysis

A Stein log-Sobolev inequality, which links KL divergence to KSD, strengthens the convergence statement. Duncan, Nüsken, and Szpruch [8] introduced this inequality and discussed conditions under which it may hold. The adaptive setting gives an important relaxation: it suffices for one kernel in the entire kernel class to satisfy the log-Sobolev inequality. We formalize this requirement as follows.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \mid \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

This property yields KL-divergence convergence bounds. In particular, it gives an explicit error bound for $\mathcal{F}(\mu_n)$ even when the maximization error is only uniformly bounded by $\varepsilon_n \leq \bar{\varepsilon}$, without requiring $\varepsilon_n \rightarrow 0$.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) - \text{KSD}_{\theta_n}^2(\mu_n \mid \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}$$

for all $n \in \mathbb{N}$.

Proof. The proof follows the same descent structure as Theorem 3. Under Assumption 5 and the stated conditions on α, γ ,

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating the recursion over $n \in \mathbb{N}$ gives the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

The three stated regimes now follow separately:

1. If $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows from the geometric-series bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1-\rho}.$$

2. If $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1-\rho/q} = q^n \frac{1}{q-\rho}$$

by the geometric-series formula. Similarly, when $q < \rho$, direct summation gives

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1-q/\rho} = \rho^n \frac{1}{(\rho-q)}.$$

When $q = \rho$, the bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ reduces the argument to the preceding cases.

3. If $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

This uses $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□